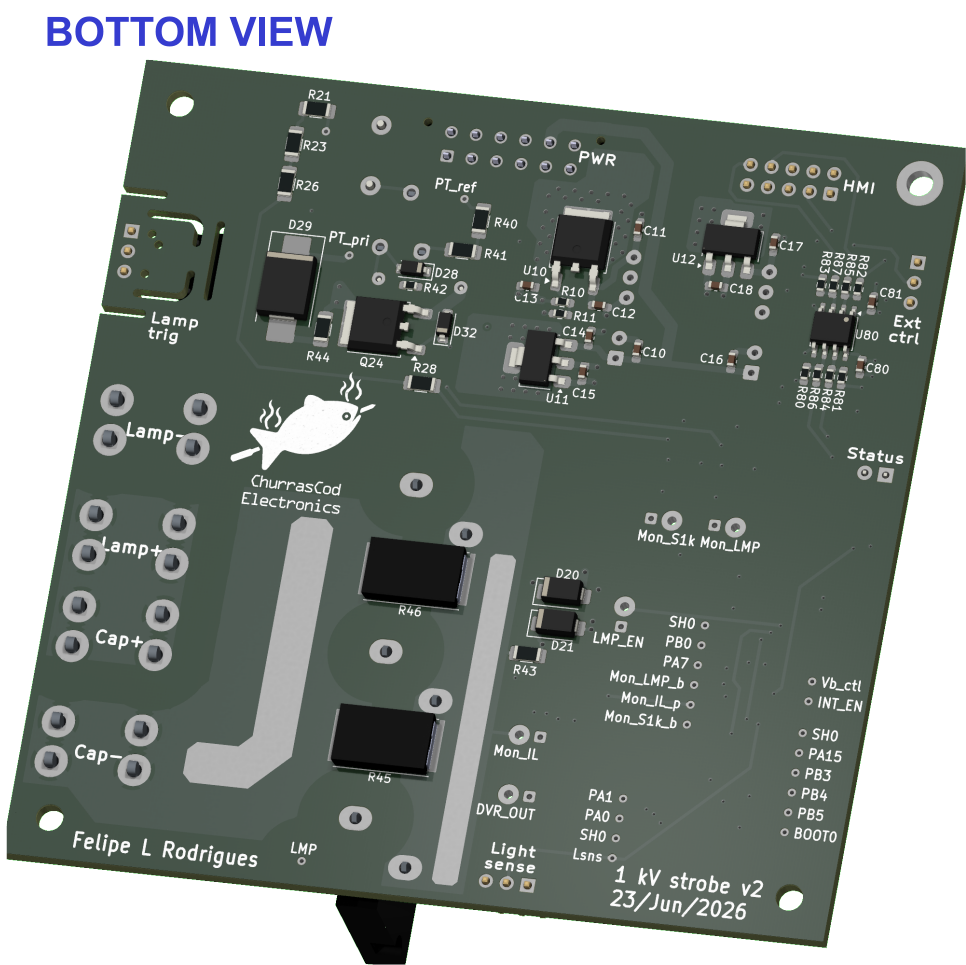
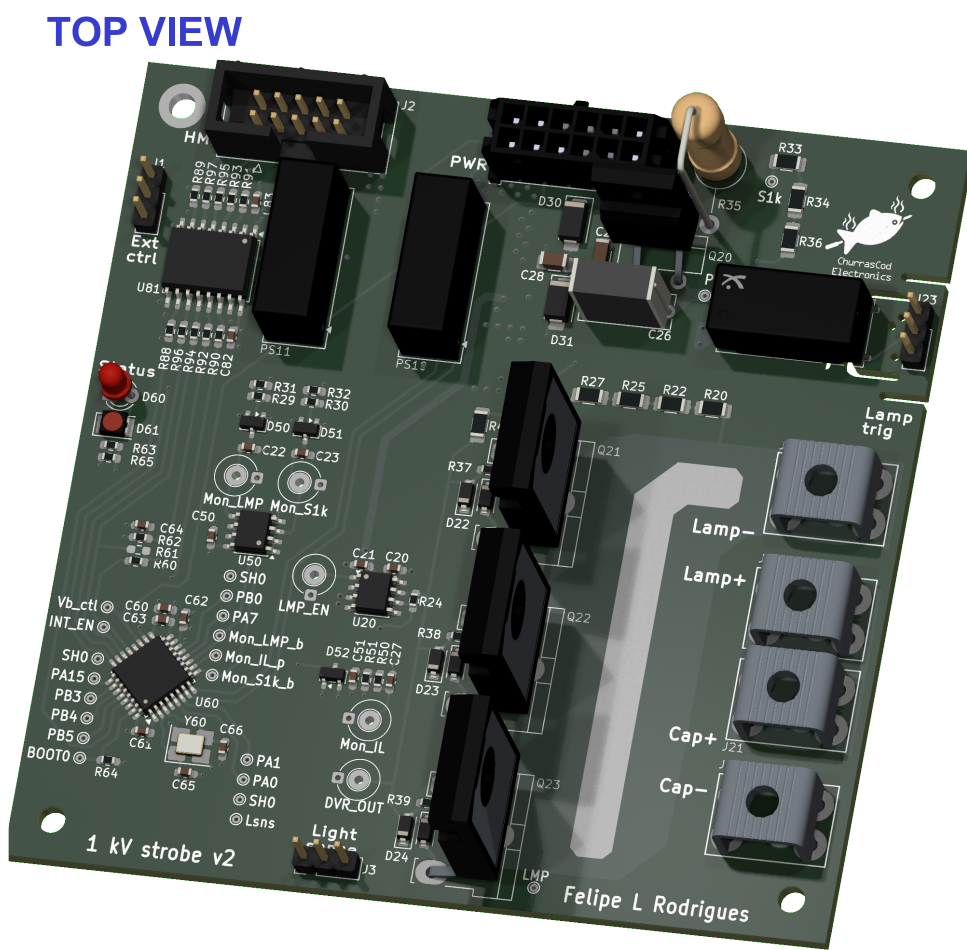


Main board

Rev 2.0.0+ (Unreleased)

2026-09-05

| Page | Index               |
|------|---------------------|
| 1    | Cover Page          |
| 2    | Block Diagram       |
| 3    | Power supply        |
| 4    | Lamp driver         |
| 5    | Signal conditioning |
| 6    | MCU                 |
| 7    | Optical isolation   |
| 8    | Revision History    |

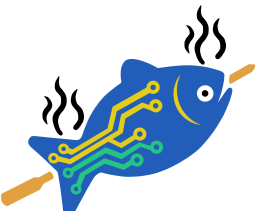


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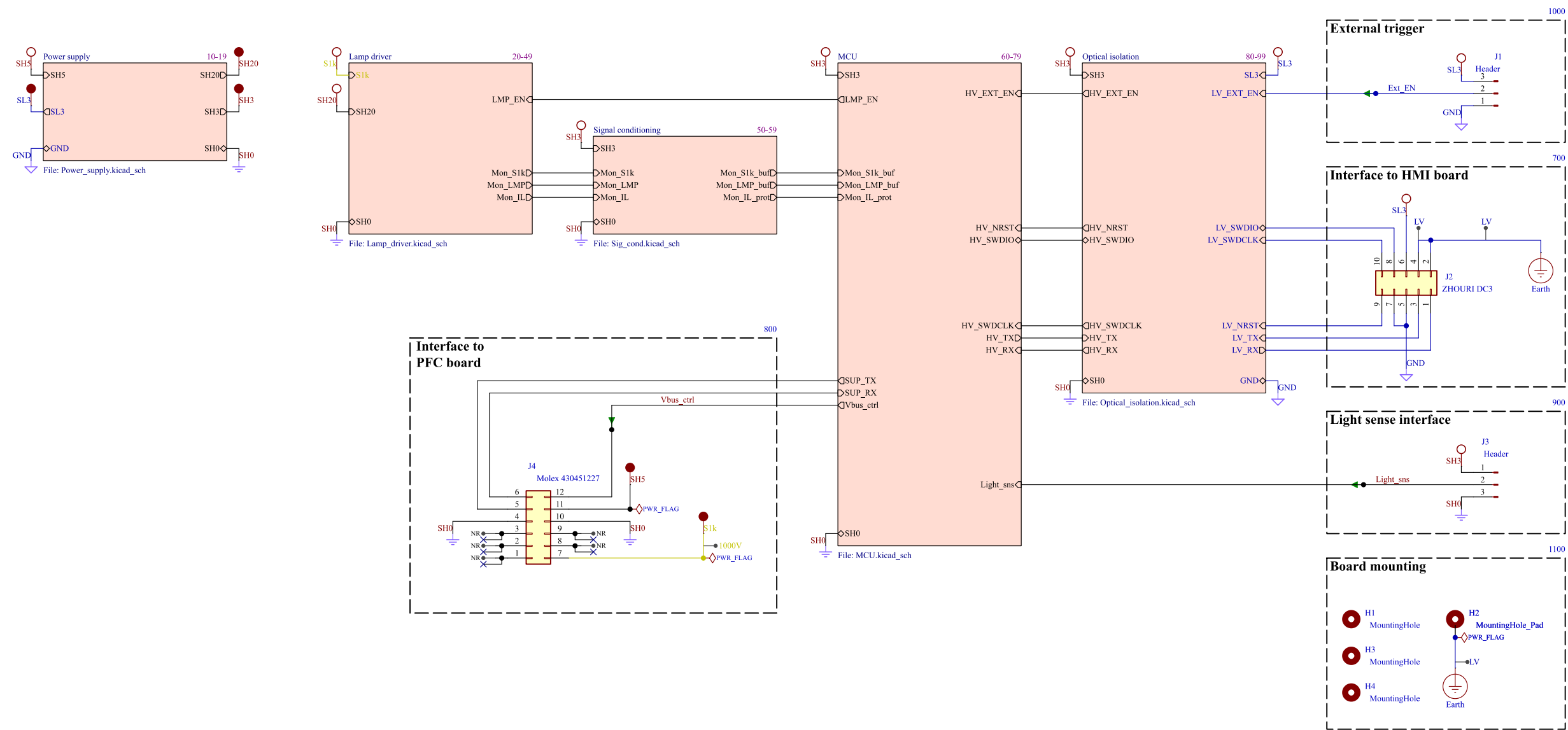
Date: 05-Sep-2026

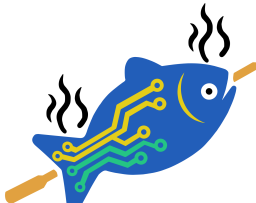
NOTE STYLING CONVENTION

|                                                                    |                                                  |                                                                 |                                                               |                                                                    |
|--------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------|
| DESIGN NOTE:<br>Example text for<br>informational<br>design notes. | DESIGN NOTE:<br>Example text for<br>debug notes. | DESIGN NOTE:<br>Example text for<br>cautionary<br>design notes. | DESIGN NOTE:<br>Example text for<br>critical design<br>notes. | LAYOUT NOTE:<br>Example text for<br>critical layout<br>guidelines. |
|--------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------|

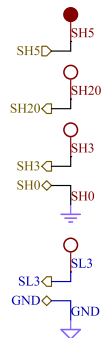
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| <br>ChurrasCod<br>Electronics | Comments:        |  | Company:<br>ChurrasCod Electronics  |                 | Variant:<br>CHECKED          | Git Hash:<br>84f4bdf             |
|                                                                                                                    |                  |  | Board Name:<br>Main board           |                 | Project Name:<br>1 kV strobe |                                  |
|                                                                                                                    | Sheet Title:     |  | File Name:<br>Strobe_main.kicad_sch | Designer:<br>FR | Date:<br>2025-07-04          | Revision:<br>2.0.0+ (Unreleased) |
|                                                                                                                    | Sheet Path:<br>/ |  | Reviewer:                           |                 | Size:<br>A3                  | Sheet:<br>1 of 8                 |

Block Diagram

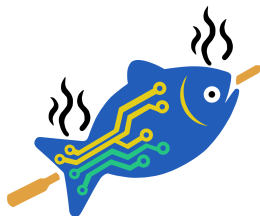
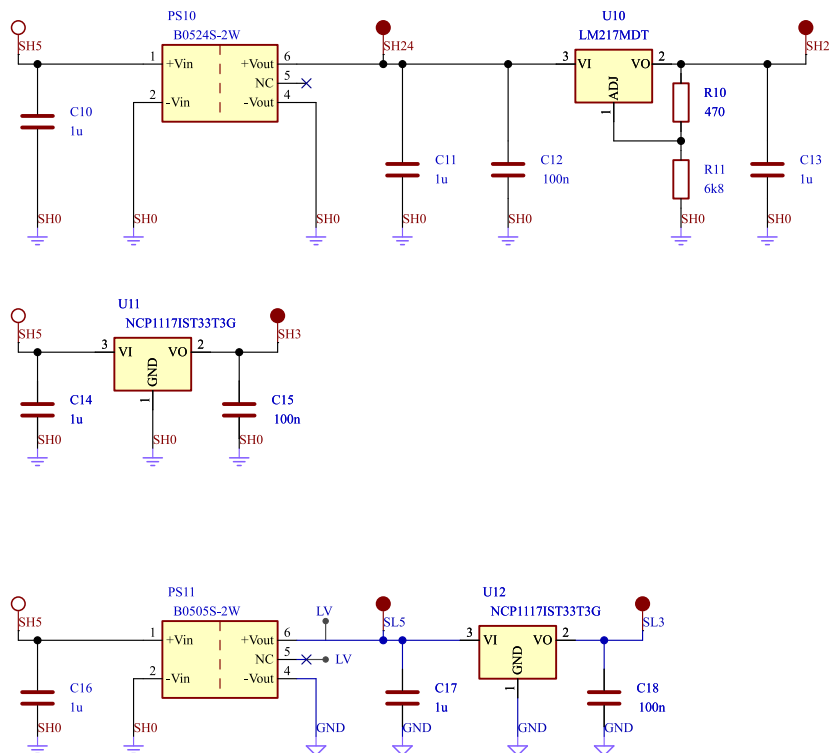


|                                                                                                                            |                                |  |                                           |  |                     |  |                                     |  |                                  |  |
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| <br><b>ChurrasCod<br/>Electronics</b> | Comments:                      |  | Company:<br><b>ChurrasCod Electronics</b> |  | Variant:<br>CHECKED |  | Git Hash:<br>84f4bdf                |  |                                  |  |
|                                                                                                                            |                                |  | Board Name:<br><b>Main board</b>          |  |                     |  | Project Name:<br><b>1 kV strobe</b> |  |                                  |  |
|                                                                                                                            | Sheet Title:<br>Block Diagram  |  | File Name:<br>Block Diagram.kicad_sch     |  | Designer:<br>FR     |  | Date:<br>2025-07-04                 |  | Revision:<br>2.0.0+ (Unreleased) |  |
|                                                                                                                            | Sheet Path:<br>/Block Diagram/ |  |                                           |  | Reviewer:           |  | Size:<br><b>A3</b>                  |  | Sheet:<br><b>2</b> of <b>8</b>   |  |

# BLOCK INTERFACE



# Power supply 10-19



ChurrasCod  
Electronics

Comments:

Sheet Title:

Power supply

Sheet Path:

/Block Diagram/Power supply/

Company:

Board Name:

Main board

File Name:

Power\_supply.kicad\_sch

Designer:

FR

Reviewer:

Variant:

CHECKED

Project Name:

1 kV strobe

Date:

2025-07-04

Size:

A4

Git Hash:

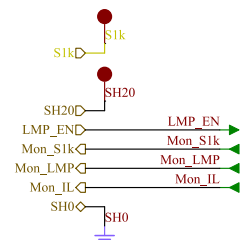
84f4bdf

Revision:

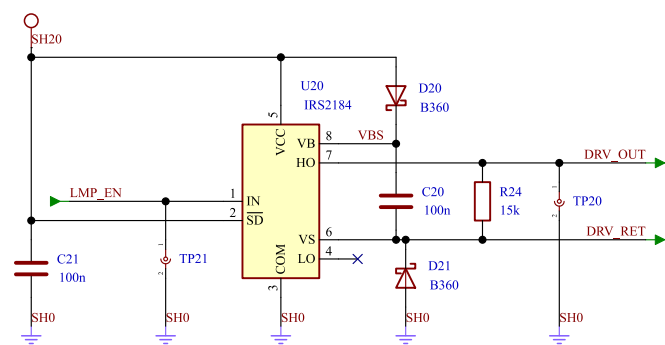
3 of 8

## Lamp driver

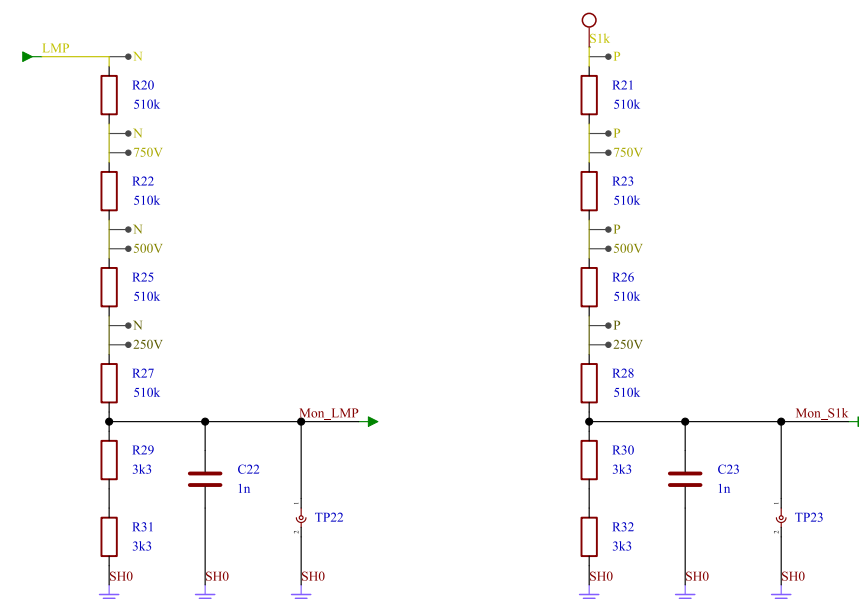
### 20-49



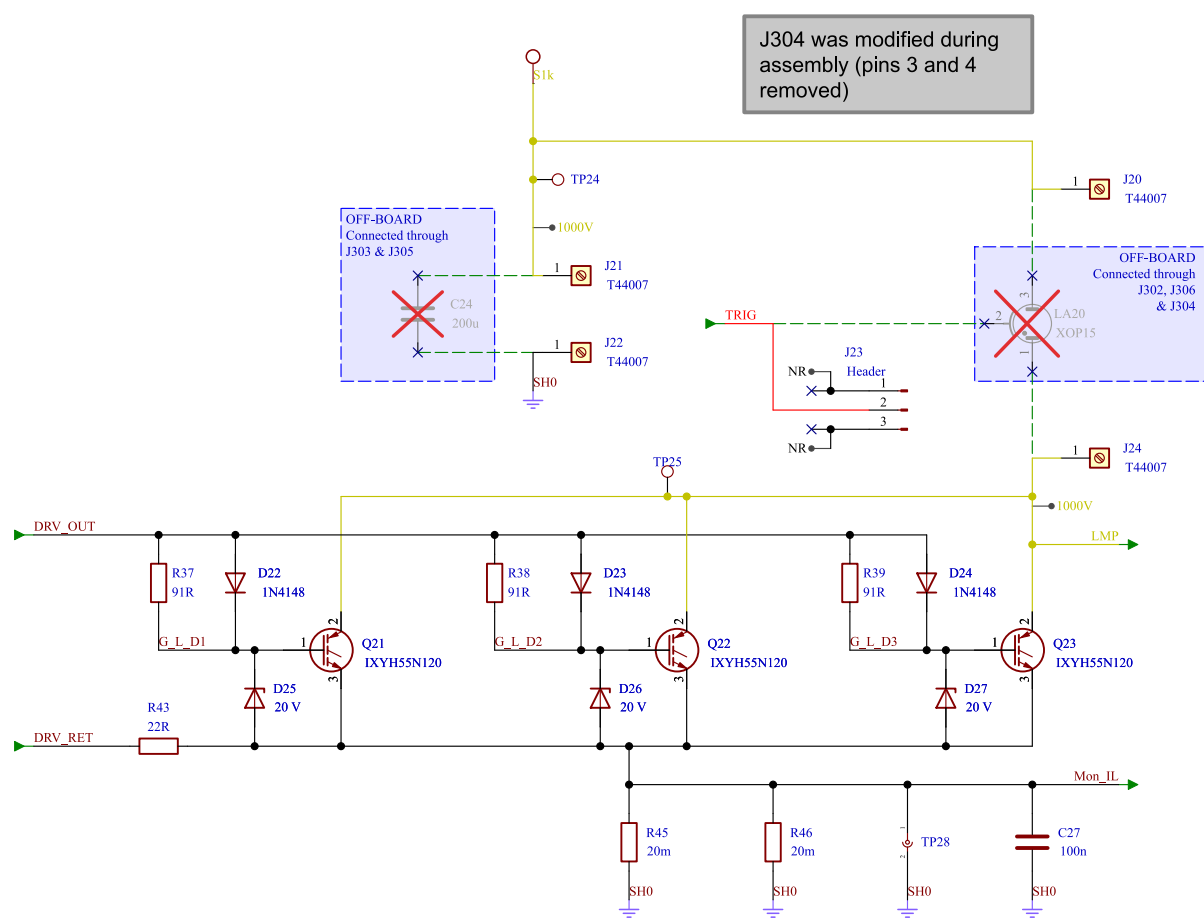
**INPUT STAGE (IGBT DRIVER)**



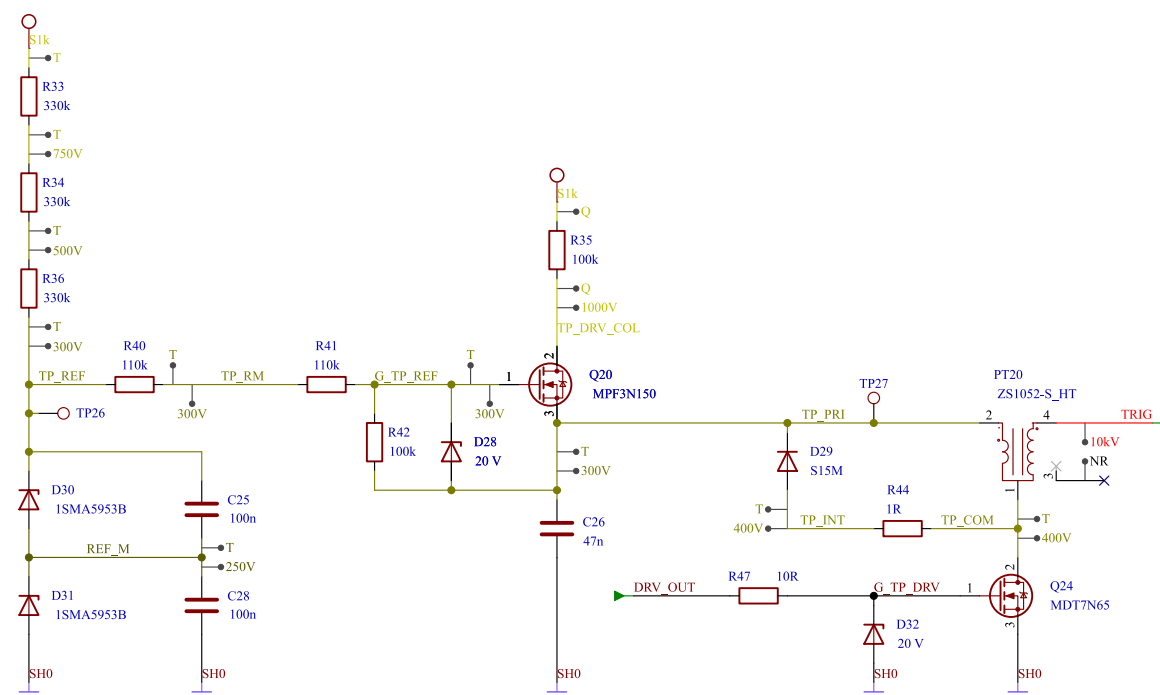
## SCALING OF HV MEASUREMENTS



**OUTPUT STAGE**

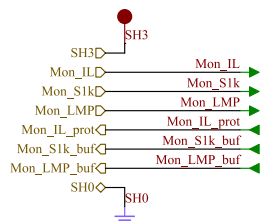


## PULSE TRANSFORMER DRIVER

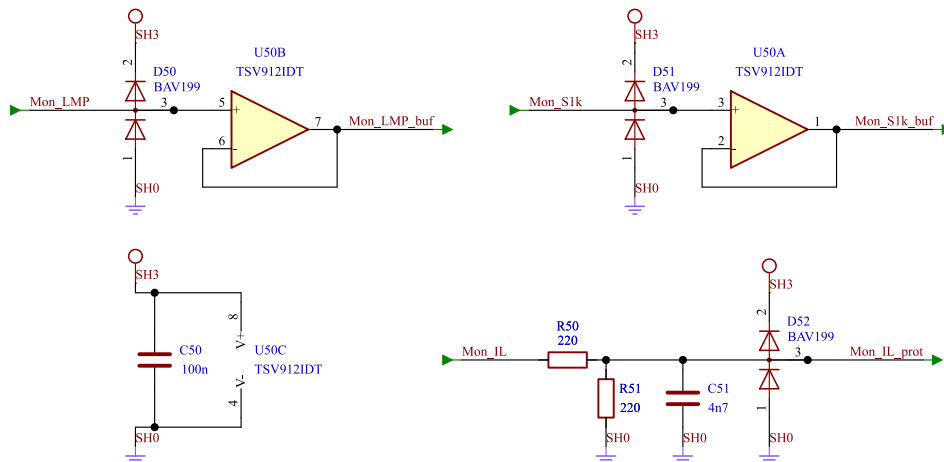


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|--------------------------------------------|-------------------------------------|-----------------|-------------------------------------|--------------------------------|
| Comments:                                  | Company:                            |                 | Variant:<br>CHECKED                 | Git Hash:<br>84f4bdf           |
|                                            | Board Name:<br><b>Main board</b>    |                 | Project Name:<br><b>1 kV strobe</b> |                                |
| Sheet Title:<br>Lamp driver                | File Name:<br>Lamp_driver.kicad_sch | Designer:<br>FR | Date:<br>2025-07-04                 | Revision:                      |
| Sheet Path:<br>/Block Diagram/Lamp driver/ |                                     | Reviewer:       | Size:<br><b>A3</b>                  | Sheet:<br><b>4</b> of <b>8</b> |

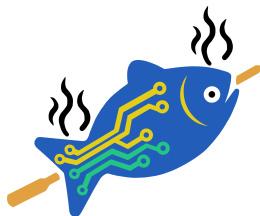
# BLOCK INTERFACE



# LAMP DRIVER VOLTAGES & CURRENT



# Signal conditioning 50-59



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Electronics

Comments:

Company:

Variant:

Git Hash:

CHECKED

84f4bdf

Board Name:

**Main board**

Project Name:

**1 kV strobe**

Sheet Title:

Signal conditioning

File Name:

Sig\_cond.kicad\_sch

Designer:

FR

Date:

2025-07-04

Revision:

Sheet Path:

/Block Diagram/Signal conditioning/

Reviewer:

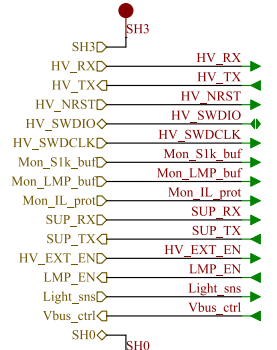
Size:

**A4**

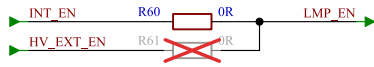
Sheet:

**5** of **8**

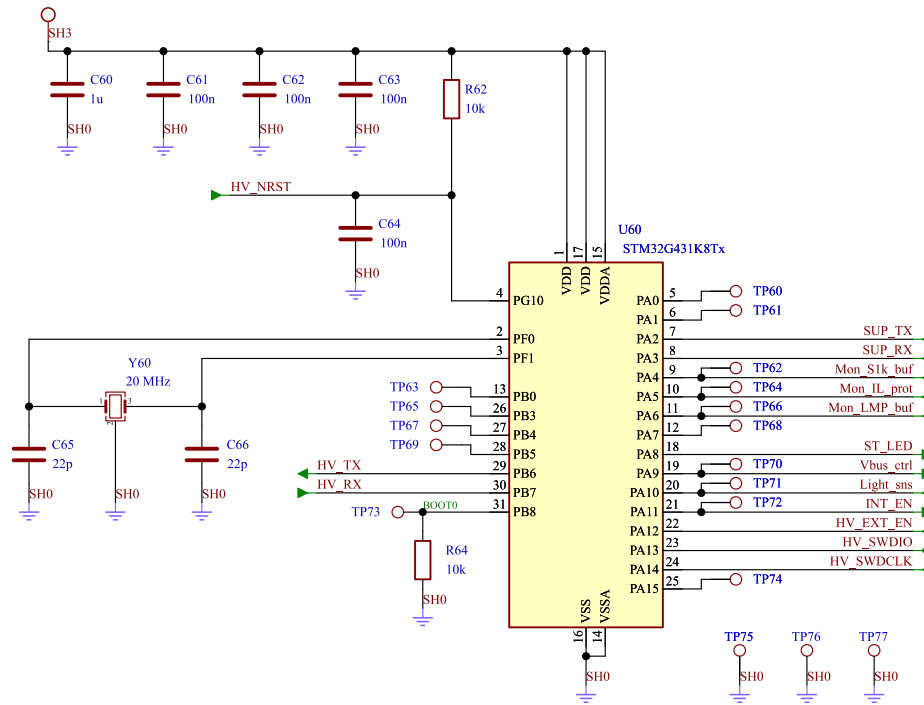
## BLOCK INTERFACE



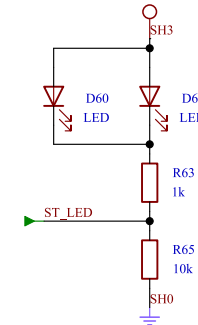
## Trigger source selection



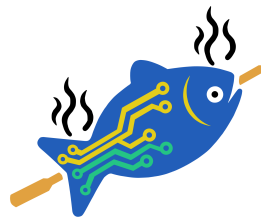
## Main controller



## Status LED



MCU  
60-79



ChurrasCod  
Electronics

Comments:

Sheet Title:

MCU

Sheet Path:

/Block Diagram/MCU/

Company:

Board Name:

Main board

File Name:

MCU.kicad\_sch

Designer:

FR

Reviewer:

Variant:

CHECKED

Project Name:

1 kV strobe

Date:

2025-07-04

Size:

A4

Git Hash:

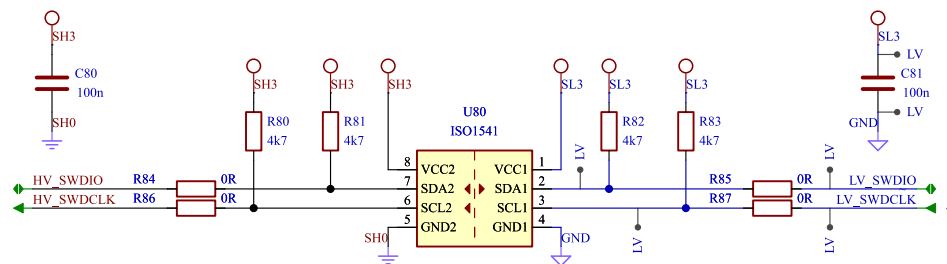
84f4bdf

Revision:

6 of 8

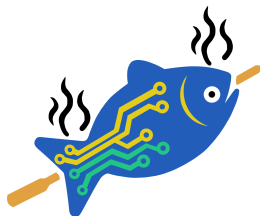
Pin connection diagram for the HV module. The diagram shows two headers with the following connections:

- SH3 (top) to HV\_SWDIO (bottom)
- SL3 (top) to HV\_SWDCLK (bottom)
- SH0 (top) to HV\_EXT\_EN (bottom)
- LV\_SWDIO (bottom) to HV\_SWDIO (top)
- LV\_SWDCLK (bottom) to HV\_SWDCLK (top)
- LV\_NRST (bottom) to HV\_NRST (top)
- LV\_TX (bottom) to HV\_TX (top)
- LV\_RX (bottom) to HV\_RX (top)
- LV\_EXT\_EN (bottom) to HV\_EXT\_EN (top)
- GND (bottom) to GND (top)



The schematic diagram illustrates the internal circuitry of the HV module, centered around the ISO6741 chip (U81). The chip is a dual-channel isolator with two input/output sections. The left section (pins 10-14) is connected to the HV module pins (HV\_RX, HV\_NRST, HV\_EXT\_EN, HV\_TX) through resistors R90, R92, R94, and R96. The right section (pins 3-7) is connected to the HV module pins (LV\_TX, LV\_NRST, LV\_EXT\_EN, LV\_RX) through resistors R91, R93, R95, and R97. The chip is powered by a 5V supply (Vcc) connected to pin 16 and a 5V supply (Vcc) connected to pin 2. The chip is also connected to ground (GND) at pins 9 and 7. The HV module pins are connected to the HV module pins through resistors R88, R89, and R98. The HV module pins are also connected to ground (GND) through capacitors C82 and C83. The HV module pins are also connected to ground (GND) through capacitors C82 and C83.

## Optical isolation



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Electronics

|                                                  |                                           |                 |                                     |                                |
|--------------------------------------------------|-------------------------------------------|-----------------|-------------------------------------|--------------------------------|
| Comments:                                        | Company:                                  |                 | Variant:<br>CHECKED                 | Git Hash:<br>84f4bdf           |
|                                                  | Board Name:<br><b>Main board</b>          |                 | Project Name:<br><b>1 kV strobe</b> |                                |
| Sheet Title:<br>Optical isolation                | File Name:<br>Optical_isolation.kicad_sch | Designer:<br>FR | Date:<br>2025-07-04                 | Revision:                      |
| Sheet Path:<br>/Block Diagram/Optical isolation/ |                                           | Reviewer:       | Size:<br><b>A4</b>                  | Sheet:<br><b>7</b> of <b>8</b> |

# Revision History

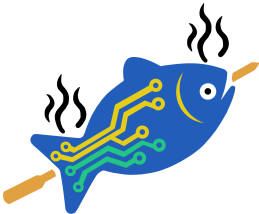
**Version 2.0.1 not found.**  
Version 2.0.1 not found.

- Version 2.0.0 - 2026-06-23**
- decreased impedance of lpeak measurement divider, removed buffer and fed signal directly to MCU
  - reworked gate drive resistors
  - reworked power input
  - modified some of the testpoints and added extra "GND testpoints"
  - reworked test pulse circuit to achive higher frequency
  - added mlcc to MCU rst line

**Version 1.0.0 - 2026-04-11**

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Migrated existing project to template (KDT)



ChurrasCod  
Electronics

|                                   |                                          |                 |                              |                                  |
|-----------------------------------|------------------------------------------|-----------------|------------------------------|----------------------------------|
| Comments:                         | Company:<br>ChurrasCod Electronics       |                 | Variant:<br>CHECKED          | Git Hash:<br>84f4bdf             |
|                                   | Board Name:<br>Main board                |                 | Project Name:<br>1 kV strobe |                                  |
| Sheet Title:<br>Revision History  | File Name:<br>Revision History.kicad_sch | Designer:<br>FR | Date:<br>2025-07-04          | Revision:<br>2.0.0+ (Unreleased) |
| Sheet Path:<br>/Revision History/ |                                          | Reviewer:       | Size:<br>A4                  | Sheet:<br>8 of 8                 |